

Name: _____

FRACTIONS (SECONDARY), (DIVIDING FRACTIONS)

Date: _____

LO: I can divide fractions by flipping the second fraction and multiplying (KFC method).

Dividing fractions

To **divide by a fraction**, **Keep** the first fraction, **Flip** the second (reciprocal), then **Change** to multiplication (KFC).

Reciprocal: The reciprocal of $\frac{a}{b}$ is $\frac{b}{a}$. Flip the fraction upside down.

KFC method

●	$\frac{2}{3} \div \frac{1}{4} = \frac{2}{3} \times \frac{4}{1} = \frac{8}{3}$	flip and multiply
●	$\frac{3}{5} \div 2 = \frac{3}{5} \times \frac{1}{2} = \frac{3}{10}$	$2 = \frac{2}{1}$
●	Reciprocal of $\frac{3}{4}$ is $\frac{4}{3}$	flip
●	$\frac{2}{3} \div \frac{1}{4} = \frac{2}{12}$	did not flip
●	Flip the first fraction	flip the SECOND

Key idea

Keep the first fraction. Flip the second. Change $\div$ to $\times$.

$$\left(\frac{a}{b}\right) \div \left(\frac{c}{d}\right) = \frac{a}{b} \times \left(\frac{d}{c}\right)$$

KFC: Keep, Flip, Change

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – basic division

Calculate $\frac{3}{4} \div \frac{1}{2}$

Keep: $\frac{3}{4}$. Flip: $\frac{1}{2}$ becomes $\frac{2}{1}$

Multiply: $\frac{3}{4} \times \frac{2}{1} = \frac{6}{4}$

Simplify: $\frac{6}{4} = \frac{3}{2} = 1\frac{1}{2}$

$$= 1\frac{1}{2}$$

Flip the second fraction then multiply across.

Example 2 – dividing by a whole number

Calculate $\frac{2}{5} \div 3$

Write 3 as $\frac{3}{1}$

Flip: $\frac{3}{1}$ becomes $\frac{1}{3}$

Multiply: $\frac{2}{5} \times \frac{1}{3} = \frac{2}{15}$

$$= \frac{2}{15}$$

A whole number $n = \frac{n}{1}$. Its reciprocal is $\frac{1}{n}$.

Example 3 – both improper

Calculate $\frac{5}{6} \div \frac{5}{3}$

Keep $\frac{5}{6}$. Flip $\frac{5}{3}$ to $\frac{3}{5}$

Multiply: $\frac{5}{6} \times \frac{3}{5} = \frac{15}{30}$

Simplify: $\frac{15}{30} = \frac{1}{2}$

$$= \frac{1}{2}$$

Cancel before multiplying if possible: $\frac{5}{6} \times \frac{3}{5} = \frac{1}{2}$.

Example 4 – mixed numbers

Calculate $2\frac{1}{2} \div 1\frac{1}{4}$

Convert: $\frac{5}{2} \div \frac{5}{4}$

Flip: $\frac{5}{2} \times \frac{4}{5}$

$= \frac{20}{10} = 2$

$$= 2$$

Convert mixed numbers to improper fractions first.

YOUR TURN – PRACTICE (deliberate practice)

1. Basic fraction division

Use KFC (Keep, Flip, Change).

a) $\frac{1}{2} \div \frac{1}{4}$

b) $\frac{3}{4} \div \frac{1}{2}$

c) $\frac{2}{3} \div \frac{1}{3}$

d) $\frac{5}{6} \div \frac{1}{3}$

2. Dividing by whole numbers

Write the whole number as a fraction first.

a) $\frac{3}{4} \div 3$

b) $\frac{2}{5} \div 4$

c) $\frac{5}{6} \div 5$

d) $\frac{7}{8} \div 2$

3. Both proper fractions

Keep, Flip, Change. Simplify your answer.

a) $\frac{2}{3} \div \frac{4}{5}$

b) $\frac{3}{8} \div \frac{3}{4}$

c) $\frac{4}{9} \div \frac{2}{3}$

d) $\frac{5}{6} \div \frac{3}{9}$

4. Mixed numbers

Convert to improper fractions first.

a) $1\frac{1}{2} \div \frac{3}{4}$

b) $2\frac{1}{3} \div \frac{1}{3}$

c) $3\frac{1}{2} \div 1\frac{3}{4}$

d) $4\frac{1}{5} \div \frac{7}{10}$

5. Mixed division problems

Calculate. Give answers in simplest form.

a) $\frac{7}{8} \div \frac{7}{4}$

b) $5 \div \frac{1}{3}$

c) $\frac{9}{10} \div \frac{3}{5}$

d) $2\frac{1}{2} \div \frac{5}{6}$

e) $\frac{1}{4} \div \frac{3}{8}$

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) $\frac{2}{3} \div \frac{1}{4} = \frac{2}{3} \times \frac{4}{1} = \frac{8}{3}$ ☐

Correct answer: _____

Reason: _____

b) Flip the first fraction when dividing ☐

Correct answer: _____

Reason: _____

c) Reciprocal of $\frac{3}{5}$ is $\frac{5}{3}$ ☐

Correct answer: _____

Reason: _____

d) $\frac{1}{2} \div \frac{1}{4} = \frac{1}{8}$ ☐

Correct answer: _____

Reason: _____

e) Dividing by $\frac{1}{2}$ is the same as multiplying by 2 ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A piece of ribbon is $3\frac{3}{4}$ metres long. It is cut into pieces of $\frac{5}{8}$ metre each. How many complete pieces can be cut? What length is left over?

Expression: _____

Simplified: _____

b) Show that dividing by $\frac{2}{3}$ is the same as multiplying by $1\frac{1}{2}$. Use this to calculate $12 \div \frac{2}{3}$ in your head.

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

